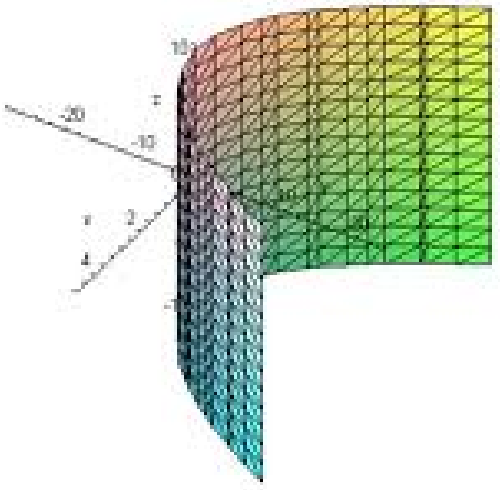
11.6 Cylinders and Quadric Surfaces

Cylinders

A *cylinder* is a surface generated by moving a straight line along a given planar curve while holding the line parallel to a given fixed line.

Can be represented by an equation missing one of the variables.

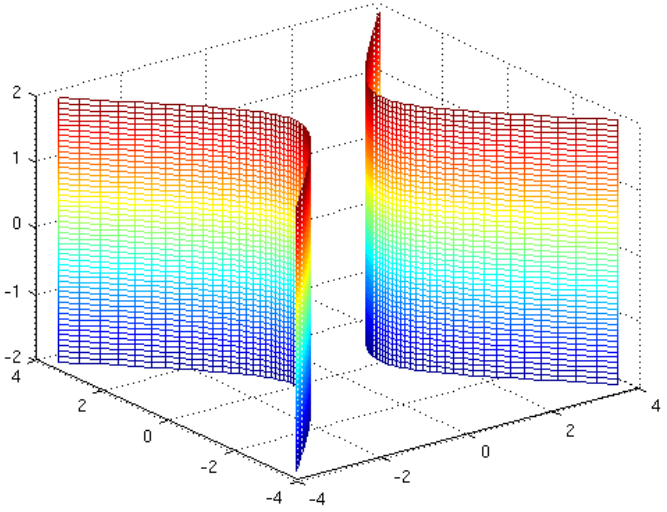
Parabolic Cylinder - eg. $y = x^2$

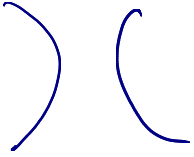


on the xy-plane, the points are $(x, x^2, 0)$.

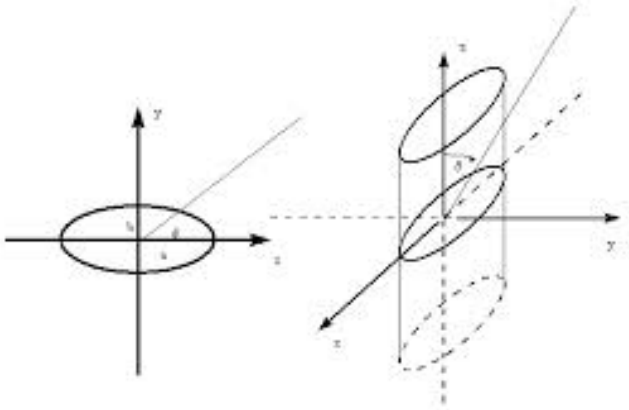
Hyperbolic Cylinder - a quadratic surface given by the equation of the form eg.

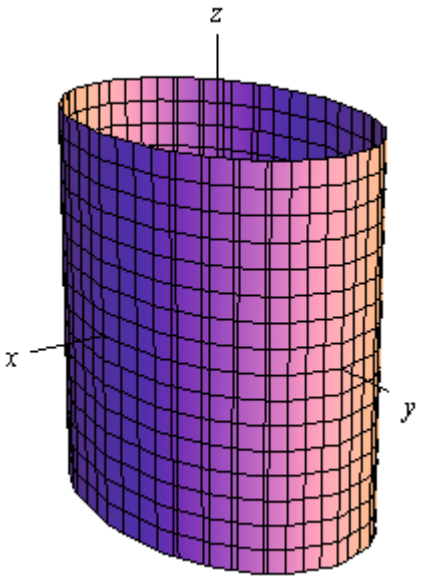
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$$





Elliptical Cylinder - A cylinder with an elliptical cross section







Quadric Surfaces

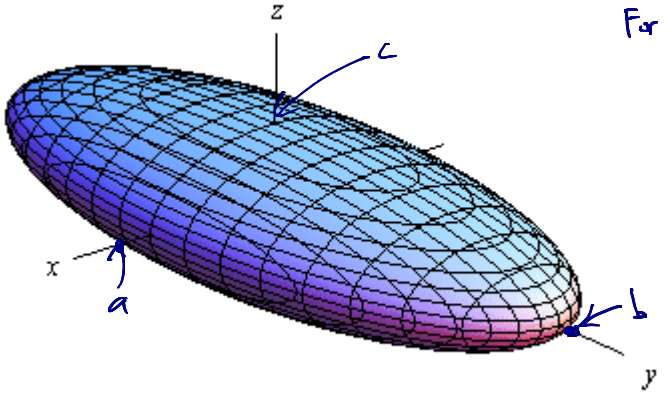
A quadric surface is the graph in space that can be put in the general form

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0,$$

where $A, ..., J$ are constants.

Note: A cylinder surface is also a quadric surface.

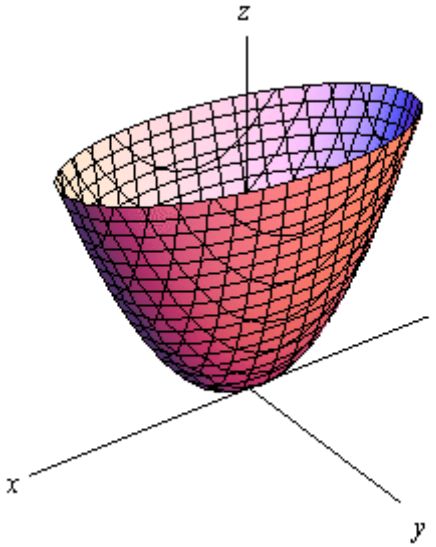
Ellipsoid - $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$



For $z = 0$, we have $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

If $a = b = c$, it is a sphere.

Elliptical Paraboloid - $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$



$z = c$

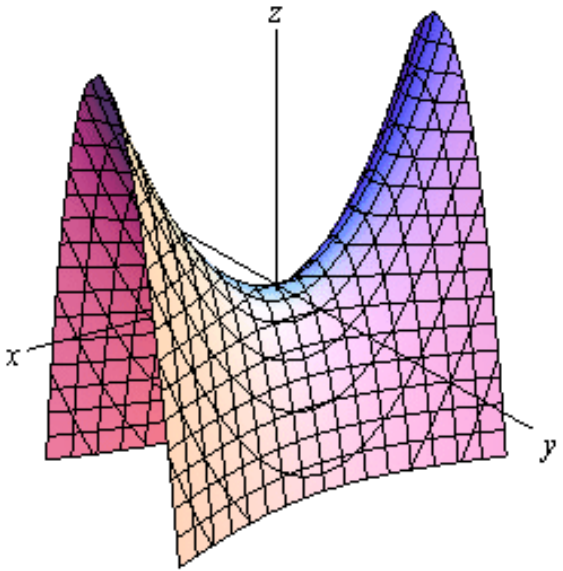
$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

When $a = b$, it will have a cross section of a circle.

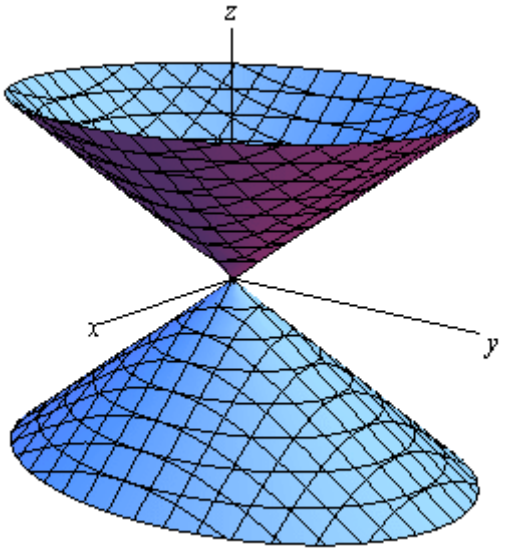
$z = 100$

$\Rightarrow \frac{x^2}{(10a)^2} + \frac{y^2}{(10b)^2} = 1$

Hyperbolic Paraboloid - $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{z}{c}$



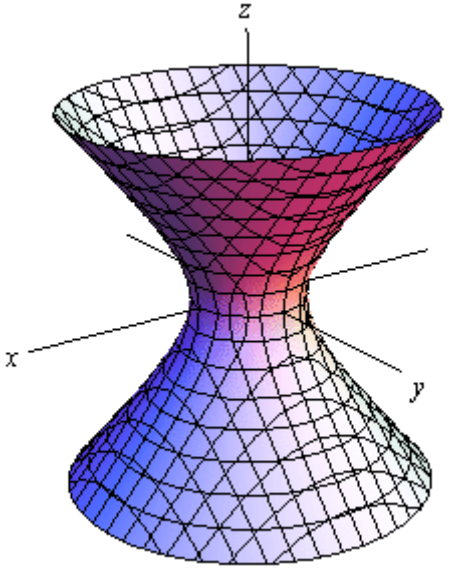
Elliptical Cone - $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$



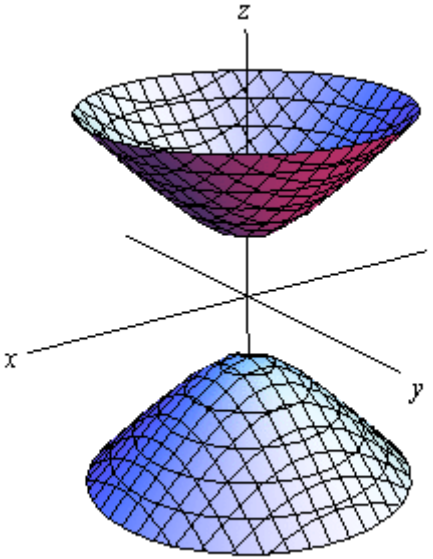
Note that this is a cone open along z-axis. Can also have cones open along x-axis (and y -axis) -

$$\frac{y^2}{b^2} + \frac{z^2}{c^2} = \frac{x^2}{a^2}$$

Hyperboloid of one sheet - $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$



Hyperboloid of two sheet - $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$



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